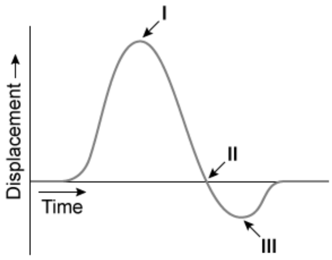
Laser Doppler vibrometers detect the Doppler shift of a laser beam reflected off an individual's chest to measure respiratory rate. The calculated displacement of the chest during a breath using a 500-nm laser is shown below.

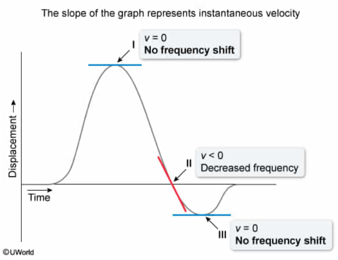


If the laser source is stationary, at which point(s) would the observed frequency of the reflected laser be 6×10^{14} Hz?

- A. II only [41%]
- B. I and II only [13%]
- C. I and III only [37%]
- D. II and III only [7%]

Correct
37% answered correctly

Explanation:



When relative motion exists between a wave source and an observer, the observed wave's frequency and wavelength differ from those of the original (Doppler effect). The Doppler shift due to the movement of the chest can be determined by comparing the frequency of the source laser to the given observed frequency. The source frequency is determined from the **wave velocity equation**; the propagation velocity *v* of a wave is the product of wavelength *λ* and frequency *f*:

$$v =$$

$$f =$$

Because a laser beam is light, *v* is equal to the speed of light $c = 3 \times 10^8$ m/s. The given laser wavelength is 500 nm, which is equal to 500×10^{-9} m. Using these values, the **source frequency** is:

$$f =$$

(see detailed calculation steps). Therefore, the given observed frequency of 6×10^{14} Hz occurs when there is **no Doppler shift**; the chest is stationary, and its velocity is zero.

Velocity is displacement divided by time, and **instantaneous velocity** is the **slope** at a given point on a displacement vs. time graph. The slope of the given graph is **zero at points I and III**, and therefore the observed frequency at these points is unchanged at 6×10^{14} Hz **(Numbers I and III)**.

(Number II) Although the *displacement* is zero at this point, the slope and relative velocity are negative; the chest is moving away. Because each wave front is reflected at a location successively farther from the source, the observed frequency of the laser will be lower than the source frequency of 6×10^{14} Hz.

Educational objective:
The frequency *f* of a wave can be determined from its propagation velocity *v* and wavelength *λ*: $f = v/λ$. If the observed frequency is the same as the source frequency, there is no frequency shift (Doppler effect). On a displacement vs. time graph, the instantaneous velocity is the slope at a given point.

Time spent:0 sec
Subject:
Physics

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System:
Light and Sound